Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/24/2007
PI Name:	Williams, James G Ph.D.		
Project Title:	Improving lunar laser ranging tests of gravitational theory		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	818/354-6466
Organization Name:	NASA Jet Propulsion Laboratory		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	10/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (1/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Turyshev, Slava (Jet Propulsion Laboratory) Boggs, Dale (Jet Propulsion Laboratory)		
Grant/Contract No.:			
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obsacurec by Earth's gravity		

Task Description:	Description: Precise analyses of accurate ranges to the Moon test several aspects of gravitational theory: the Equivalence Principle, geodetic precession, PPN parameters beta and gamma, and possible variation of the gravitational constant G. The uncertainties of these tests decrease as data accuracies improve and data time span lengthens. This project seeks to improve the Lunar Laser Ranging (LLR) models to exploit the University of California at San Diego/University of Washington APOLLO effort's mm accuracies for improved tests of fundamental and gravitational physics. The goals of this project are three-fold: 1) Develop and implement the multidisciplinary modeling and data analysis improvements necessary to advance LLR from the current cm range accuracy to the expected mm accuracy of the APOLLO effort; 2) Support this new ranging effort, including tests for data accuracy and short- and long-term stability; 3) For future space exploration, investigate the performance characteristics which are desirable for the next generation of laser ranging, the fundamental questions which can be investigated, and the challenges for modeling and data analysis. Objective: Precise ranges to the Moon must be accurately analyzed in order to extract the scientific information on the nature of gravity. Accurate models for the computation of the orbit of the Moon and the rotation and orientation of the Earth and Moon, and their realization in computer programs, are needed to do this. This task improves the models and computer programs used for analyzing the Lunar Laser Ranging data. The intent is to improve the accuracy of the data analysis by an order of magnitude in order to match the mm accuracy of the data expected from the new APOLLO facility. The analysis of the highly accurate lunar range data would be used to study the nature of gravitation. Significance: Einstein's theory of gravity has been tested by Lunar Laser Ranging and other techniques. This task would improve the accuracy of LLR tests by an order of magnitude. Physics is seeking the next theory of gravity beyond Einstein's theory. The accuracy of the range data is being improved to provide more accurate data on the nature of gravity. An improvement in data accuracy requires an improvement in analysis accuracy if this exploration of gravitation is to be realized.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Fundamental research on gravitational physics aids the search for new physics. Ephemeris of Moon is publicly available and is used by space missions. Data is also sensitive to geophysical/geodetic parameters.
Task Progress:	In summary for 2004-2007: a) A paper on model improvements is published, a paper on LLR tests of the equivalence principle remains in press and two papers with J. Mueller are published. A paper on ranging to the Moon and beyond has been accepted for publication. A chapter was submitted and accepted for the Explanatory Supplement to the Astronautical Almanac, but that book is progressing slowly if at all. Several presentations were given by S. G. Turyshev and one by J. G. Williams. See the bibliography for paper and abstract references. b) Following the checkout phase at Apache Point, there are 147 accurate ranges from April 2006 to May 2007 which we fit. Residuals were sent to Tom Murphy (UCSD). c) Three improvements were made to the model for integration of lunar orbit and rotation. Two were internal definition changes and the third added Mars torques. Parallel changes were made in integrated partial derivatives programs. d) A new finite difference partial derivatives program was constructed. New partial derivatives files were made. e) A change of relativistic scale for Earth and Moon sites was made in our range processing program. The constant part of the scale change can be viewed as a change of definition, but there is also a 1 mm annual effect. f) Code was modified to add new stations. Station coordinates and coordinate rates can be fit. Apache Point and Matera are the new LLR stations added. g) The LLR integrator code for asteroid perturbations was extensively modified. h) The first jointly fit lunar and planetary ephemeris in a decade has been prepared. It is designated DE418. To achieve this, comparisons between lunar and planetary software were made, compatibilities were promoted, and the above integrator asteroid modifications were made. i) A shift of computer platforms is necessary and supporting modifications to code have been started. This has been a time consuming project. A description of progress with greater detail can be found in the final report.
Bibliography Type:	Description: (Last Updated: 10/31/2007)
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Articles in Peer-reviewed Journals	Williams JG, Turyshev SG, Murphy TW. "Improving LLR tests of gravitational theory." International Journal of Modern Physics D. 2004 Mar;13(3):567-82. http://dx.doi.org/10.1142/S0218271804004682 , Mar-2004
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